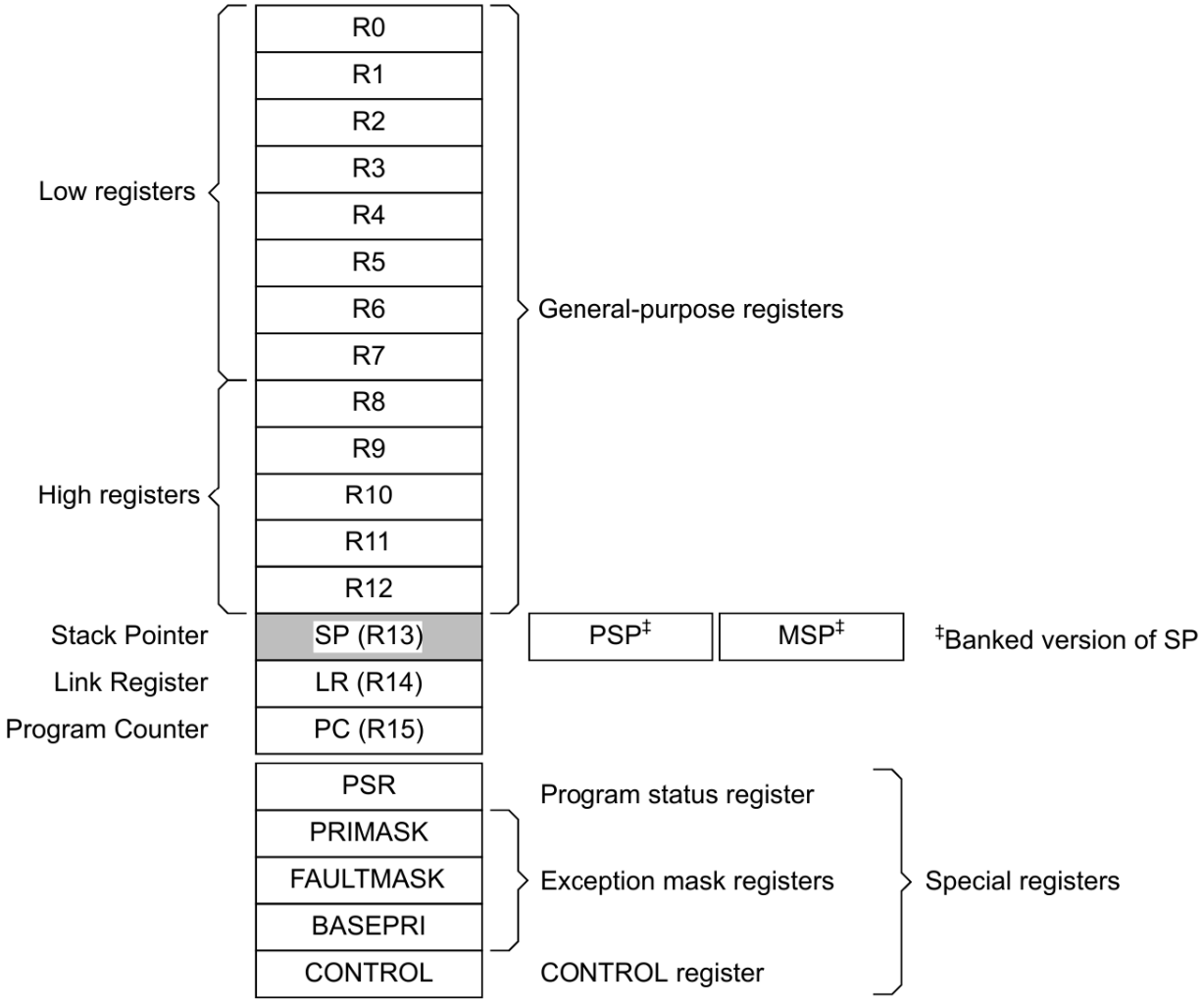
Register sets

Location: 2.3.3. Register Map

Figure 2-3. Cortex-M4F Register Set



Register 60: General-Purpose Input/Output Run Mode Clock Gating Control (RCGCGPIO), offset 0x608

General-Purpose Input/Output Run Mode Clock Gating Control (RCGCGPIO)

Base 0x400F.E000
Offset 0x608
Type RW, reset 0x0000.0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
reserved															
Type	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
reserved										R5	R4	R3	R2	R1	R0
Type	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RW	RW	RW	RW	RW	RW
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

The **RCGCGPIO** register provides software the capability to enable and disable GPIO modules in Run mode. When enabled, a module is provided a clock and accesses to module registers are allowed. When disabled, the clock is disabled to save power and accesses to module registers generate a bus fault. This register provides the same capability as the legacy **Run Mode Clock Gating Control Register n RCGCn** registers specifically for the watchdog modules and has the same bit polarity as the corresponding **RCGCn** bits.

Important: This register should be used to control the clocking for the GPIO modules. To support legacy software, the **RCGC2** register is available. A write to the **RCGC2** register also writes the corresponding bit in this register. Any bits that are changed by writing to the **RCGC2** register can be read back correctly with a read of the **RCGC2** register. Software must use this register to support modules that are not present in the legacy registers. If software uses this register to write a legacy peripheral (such as GPIO A), the write causes proper operation, but the value of that bit is not reflected in the **RCGC2** register. If software uses both legacy and peripheral-specific register accesses, the peripheral-specific registers must be accessed by read-modify-write operations that affect only peripherals that are not present in the legacy registers. In this manner, both the peripheral-specific and legacy registers have coherent information.

Bit/Field	Name	Type	Reset	Description
31:6	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
5	R5	RW	0	GPIO Port F Run Mode Clock Gating Control

Bit/Field	Name	Type	Reset	Description
				Value Description 0 GPIO Port F is disabled. 1 Enable and provide a clock to GPIO Port F in Run mode.
4	R4	RW	0	GPIO Port E Run Mode Clock Gating Control Value Description 0 GPIO Port E is disabled. 1 Enable and provide a clock to GPIO Port E in Run mode.
3	R3	RW	0	GPIO Port D Run Mode Clock Gating Control Value Description 0 GPIO Port D is disabled. 1 Enable and provide a clock to GPIO Port D in Run mode.
2	R2	RW	0	GPIO Port C Run Mode Clock Gating Control Value Description 0 GPIO Port C is disabled. 1 Enable and provide a clock to GPIO Port C in Run mode.
1	R1	RW	0	GPIO Port B Run Mode Clock Gating Control Value Description 0 GPIO Port B is disabled. 1 Enable and provide a clock to GPIO Port B in Run mode.
0	R0	RW	0	GPIO Port A Run Mode Clock Gating Control Value Description 0 GPIO Port A is disabled. 1 Enable and provide a clock to GPIO Port A in Run mode.

Register 18: GPIO Digital Enable (GPIODEN), offset 0x51C

Note: Pins configured as digital inputs are Schmitt-triggered.

The **GPIODEN** register is the digital enable register. By default, all GPIO signals except those listed below are configured out of reset to be undriven (tristate). Their digital function is disabled; they do not drive a logic value on the pin and they do not allow the pin voltage into the GPIO receiver. To use the pin as a digital input or output (either GPIO or alternate function), the corresponding GPIODEN bit must be set.

Important: The table below shows special consideration GPIO pins. Most GPIO pins are configured as GPIOs and tri-stated by default (**GPIOAFSEL**=0, **GPIODEN**=0, **GPIOPDR**=0, **GPIOPUR**=0, and **GPIOPCTL**=0). Special consideration pins may be programed to a non-GPIO function or may have special commit controls out of reset. In addition, a Power-On-Reset (POR) or asserting RST returns these GPIO to their original special consideration state.

Table 10-10. GPIO Pins With Special Considerations

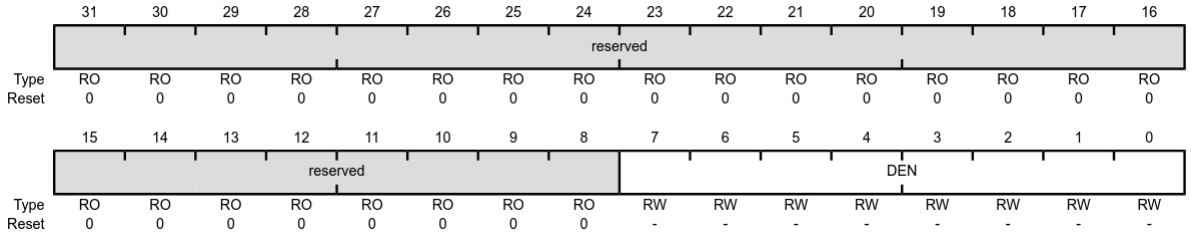
GPIO Pins	Default Reset State	GPIOAFSEL	GPIODEN	GPIODR	GPIOPUR	GPIOPCTL	GPIOCR
PA[1:0]	UART0	0	0	0	0	0x1	1
PA[5:2]	SSI0	0	0	0	0	0x2	1
PB[3:2]	I²C0	0	0	0	0	0x3	1
PC[3:0]	JTAG/SWD	1	1	0	1	0x1	0
PD[7]	GPIO[a]	0	0	0	0	0x0	0
PF[0]	GPIO[a]	0	0	0	0	0x0	0

a. This pin is configured as a GPIO by default but is locked and can only be reprogrammed by unlocking the pin in the **GPIOLOCK** register and uncommitting it by setting the **GPIOCR** register.

The GPIO commit control registers provide a layer of protection against accidental programming of critical hardware signals including the GPIO pins that can function as JTAG/SWD signals and the NMI signal. The commit control process must be followed for these pins, even if they are programmed as alternate functions other than JTAG/SWD or NMI; see “Commit Control” on page 656.

Note: The GPIO commit control registers provide a layer of protection against accidental programming of critical hardware peripherals. Protection is provided for the GPIO pins that can be used as the four JTAG/SWD pins and the NMI pin (see “Signal Tables” on page 1329 for pin numbers). Writes to protected bits of the **GPIO Alternate Function Select (GPIOAFSEL)** register (see page 671), **GPIO Pull Up Select (GPIOPUR)** register (see page 677), **GPIO Pull-Down Select (GPIOPDR)** register (see page 679), and **GPIO Digital Enable (GPIODEN)** register (see page 682) are not committed to storage unless the **GPIO Lock (GPIOLOCK)** register (see page 684) has been unlocked and the appropriate bits of the **GPIO Commit (GPIOCR)** register (see page 685) have been set.

GPIO Digital Enable (GPIODEN)
GPIO Port A (APB) base: 0x4000.4000
GPIO Port A (AHB) base: 0x4005.8000
GPIO Port B (APB) base: 0x4000.5000
GPIO Port B (AHB) base: 0x4005.9000
GPIO Port C (APB) base: 0x4000.6000
GPIO Port C (AHB) base: 0x4005.A000
GPIO Port D (APB) base: 0x4000.7000
GPIO Port D (AHB) base: 0x4005.B000
GPIO Port E (APB) base: 0x4002.4000
GPIO Port E (AHB) base: 0x4005.C000
GPIO Port F (APB) base: 0x4002.5000
GPIO Port F (AHB) base: 0x4005.D000
Offset 0x51C
Type RW, reset -

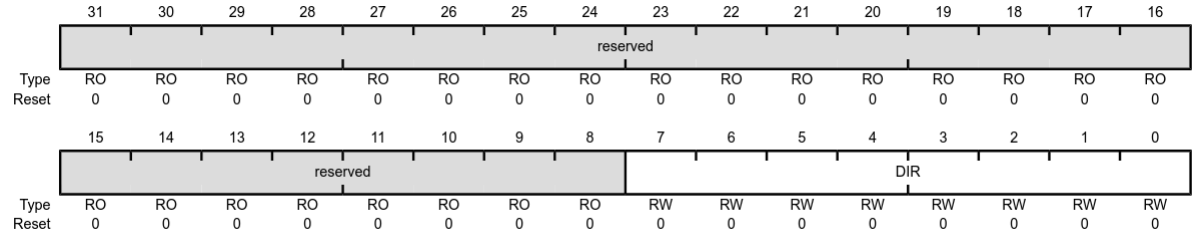


Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x0000.00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	DEN	RW	-	Digital Enable Value Description 0 The digital functions for the corresponding pin are disabled. 1 The digital functions for the corresponding pin are enabled. The reset value for this register is 0x0000.0000 for GPIO ports that are not listed in Table 10-1 on page 650.

Register 2: GPIO Direction (GPIODIR), offset 0x400

The GPIODIR register is the data direction register. Setting a bit in the GPIODIR register configures the corresponding pin to be an output, while clearing a bit configures the corresponding pin to be an input. All bits are cleared by a reset, meaning all GPIO pins are inputs by default.

GPIO Direction (GPIODIR)
GPIO Port A (APB) base: 0x4000.4000
GPIO Port A (AHB) base: 0x4005.8000
GPIO Port B (APB) base: 0x4000.5000
GPIO Port B (AHB) base: 0x4005.9000
GPIO Port C (APB) base: 0x4000.6000
GPIO Port C (AHB) base: 0x4005.A000
GPIO Port D (APB) base: 0x4000.7000
GPIO Port D (AHB) base: 0x4005.B000
GPIO Port E (APB) base: 0x4002.4000
GPIO Port E (AHB) base: 0x4005.C000
GPIO Port F (APB) base: 0x4002.5000
GPIO Port F (AHB) base: 0x4005.D000
Offset 0x400
Type RW, reset 0x0000.0000



Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x0000.00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	DIR	RW	0x00	GPIO Data Direction Value Description 0 Corresponding pin is an input. 1 Corresponding pins is an output.

Register 1: GPIO Data (GPIODATA), offset 0x000

The **GPIODATA** register is the data register. In software control mode, values written in the **GPIODATA** register are transferred onto the GPIO port pins if the respective pins have been configured as outputs through the **GPIO Direction (GPIODIR)** register (see page 663).

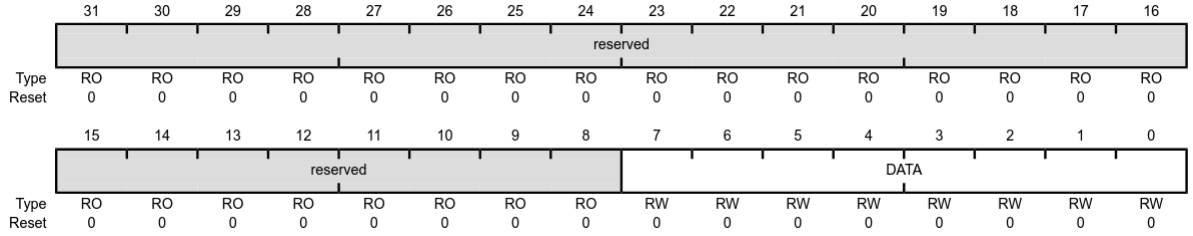
In order to write to **GPIODATA**, the corresponding bits in the mask, resulting from the address bus bits `[9:2]`, must be set. Otherwise, the bit values remain unchanged by the write.

Similarly, the values read from this register are determined for each bit by the mask bit derived from the address used to access the data register, bits `[9:2]`. Bits that are set in the address mask cause the corresponding bits in **GPIODATA** to be read, and bits that are clear in the address mask cause the corresponding bits in **GPIODATA** to be read as 0, regardless of their value.

A read from **GPIODATA** returns the last bit value written if the respective pins are configured as outputs, or it returns the value on the corresponding input pin when these are configured as inputs. All bits are cleared by a reset.

GPIO Data (GPIODATA)
GPIO Port A (APB) base: 0x4000.4000
GPIO Port A (AHB) base: 0x4005.8000
GPIO Port B (APB) base: 0x4000.5000
GPIO Port B (AHB) base: 0x4005.9000
GPIO Port C (APB) base: 0x4000.6000
GPIO Port C (AHB) base: 0x4005.A000
GPIO Port D (APB) base: 0x4000.7000
GPIO Port D (AHB) base: 0x4005.B000
GPIO Port E (APB) base: 0x4002.4000

GPIO Port E (AHB) base: 0x4005.C000
GPIO Port F (APB) base: 0x4002.5000
GPIO Port F (AHB) base: 0x4005.D000
Offset 0x000
Type RW, reset 0x0000.0000



Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x0000.00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	DATA	RW	0x00	GPIO Data This register is virtually mapped to 256 locations in the address space. To facilitate the reading and writing of data to these registers by independent drivers, the data read from and written to the registers are masked by the eight address lines [9:2]. Reads from this register return its current state. Writes to this register only affect bits that are not masked by ADDR[9:2] and are configured as outputs. See “Data Register Operation” on page 654 for examples of reads and writes.

Data Register Operation

To aid in the efficiency of software, the GPIO ports allow for the modification of individual bits in the **GPIO Data (GPIODATA)** register (see page 662) by using bits `[9:2]` of the address bus as a mask. In this manner, software drivers can modify individual GPIO pins in a single instruction without affecting the state of the other pins. This method is more efficient than the conventional method of performing a read-modify-write operation to set or clear an individual GPIO pin. To implement this feature, the **GPIODATA** register covers 256 locations in the memory map.

During a write, if the address bit associated with that data bit is set, the value of the **GPIODATA** register is altered. If the address bit is cleared, the data bit is left unchanged.

For example, writing a value of `0xEB` to the address GPIODATA + `0x098` has the results shown in Figure 10-3, where `u` indicates that data is unchanged by the write. This example demonstrates how **GPIODATA** bits 5, 2, and 1 are written.

Figure 10-3. GPIODATA Write Example

ADDR[9:2]	9	8	7	6	5	4	3	2	1	0
0x098	0	0	1	0	0	1	1	0	0	0
0xEB	**1**	**1**	**1**	**0**	**1**	**0**	**1**	**1**		
GPIODATA	**u**	**u**	**1**	**u**	**u**	**0**	**1**	**u**		
	7	6	5	4	3	2	1	0		

During a read, if the address bit associated with the data bit is set, the value is read. If the address bit associated with the data bit is cleared, the data bit is read as a zero, regardless of its actual value. For example, reading address GPIODATA + 0x0C4 yields as shown in Figure 10-4. This example shows how to read **GPIODATA** bits 5, 4, and 0.

Figure 10-4. GPIODATA Read Example

ADDR[9:2]	9	8	7	6	5	4	3	2	1	0
0x0C4	0	0	1	1	0	0	0	1	0	0
GPIODATA	**1**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**0**
Returned Value	**0**	**0**	**1**	**1**	**0**	**0**	**0**	**0**		
	7	6	5	4	3	2	1	0		